any other currency, cryptocurrencies can also be used to buy goods and services. You can also trade them for profit. Based on the technology of 'blockchain,' their decentralised nature makes them theoretically immune to the old ways of government control and interference.

Blockchain is a continuously growing list of records called blocks, which are linked and secured using cryptography. It is an open, distributed ledger that can record transactions between two parties efficiently and in a verifiable and permanent way. For use as a distributed ledger, the blockchain is managed by a peer-to-peer network collectively adhering to a protocol for validating new blocks. Once recorded, the data in any given block cannot be altered retroactively.

Background

Filing for bankruptcy in September 2008 by Lehman Brothers of the US nucleated an unprecedented global financial crisis. This spurred initiatives for an alternative.

Already working on blockchain, a talented computer coder with pseudoname Satoshi Nakamoto proposed Bitcoin on January 3, 2009, as a secure digital currency. There was no paper, copper, or silver—just 31,000 lines of codes and an announcement on the Internet.

Today, it is legal in the USA, Japan, South Korea, Russia, the UK, and Brazil besides European countries like Italy, Sweden, and the Netherlands. Globally, over 2000 cryptocurrencies exist as of January 2020. Bitcoin service charges are usually much less than credit cards.

Central bankers like RBI can print any amount of national currency, causing inflation. However, the 'finite' availability of cryptocurrencies makes them deflationary. For example, Bitcoin has a programmed supply limit of 21 million units. This feature is similar to a precious metal like gold, which is limited in supply on Earth.

Working of cryptocurrency

Cryptocurrencies are developed as code by teams who build in mechanisms for issuance. Cryptographic protocols developed on advanced mathematics and computer engineering principles make it virtually impossible to break, duplicate, or counterfeit the protected currencies. Identities of users are secret, making transactions and fund flows difficult to attribute to specific individuals or groups.

Using innovative cryptography, the Bitcoin software encrypts each transaction—the sender and receiver are identified only by a string of numbers—but a public record of every coin's movement is published across the entire network. Buyers and sellers remain anonymous, but everyone can see that a coin has moved from A to B, and Nakamoto's code can prevent A from spending the coin a second time. Cryptocurrency coins are generated through a process called mining.

The software allows people to send money directly to each other without an intermediary, and no outside party can create more Bitcoins. Central banks and governments play no role. "Everything is based on crypto proof instead of trust," Nakamoto wrote in his 2009 essay.

Dan Kaminsky, a leading Internet-security researcher, comments, "He's a world-class programmer, with a deep understanding of the C++ programming language. He understands economics, cryptography, and peer-to-peer networking."

People who own Bitcoins have a program called Bitcoin client installed on their computers to manage their accounts. When they want to access their funds, they use the client to send a transaction request.

Transactions don't require you to give up any secret information. Instead, they use two keys—a public key and a private key. The public key can be seen by everyone (it is actually your bitcoin address), but your private key is secret.

When you send a bitcoin, you 'sign' the transaction by combining your public and private keys and applying a mathematical function to them. This creates a certificate that proves the transaction came from you. Also, it protects merchants from losses caused by fraud or fraudulent chargebacks.

Cryptocurrency ATM (Credit: https://upload.wikimedia.org)

A cryptocurrency wallet stores the public and private keys or addresses, which can be used to receive or spend the cryptocurrency.

Altcoins

Currencies modelled after Bitcoin are collectively called altcoins. Ten prominent ones are—Ethereum, Ripple, Litecoin, Tether, BCH, Libra, Monero, EOS, BSV, and BNB.

For instance, launched in 2015, Ethereum is positioned as a toolkit for financial innovation and experimentation. Launched in 2012, Ripple reduces the usage of computing power and minimises network latency. Created by Charlie Lee, an MIT graduate, Litecoin features a faster block generation rate. Launched in 2014, Tether minimises volatility. This year, Facebook is launching its cyber currency, Libra.

Environmental impact

Depending on the energy source, researchers estimate that cryptomining can produce three to fifteen million tons of global carbon emissions. Cryptocurrency 'miners' produce currency through energy-intensive 'mining' processes, requiring extensive computing resources.

By 2018-end, Bitcoin mining farms had consumed 0.05 per cent of the world's energy. These energy consumption levels were equivalent to the net power consumption of nations such as Ireland. China is the dominant force in the cryptocurrency industry, harbouring the world's largest Bitcoin mining companies while simultaneously facilitating the most Bitcoin transaction traffic.

To lower operating costs, miners